The CHAPTER GRIB2 archive

A 3 km dynamical downscaling of ERA5
Content and mapping onto the ERA5 schema

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Contents

1 The simulation **1**
- 1.1 Domain, grid and period . . . 1
- 1.2 Physical configuration . . . 1
- 1.3 The static file . . . 2

2 The archive format **3**
- 2.1 Files and messages . . . 3
- 2.2 Geometry . . . 3
- 2.3 Vertical levels . . . 3
- 2.4 Time convention . . . 4
- 2.5 The run switch . . . 4

3 Rules of correspondence with ERA5 **6**
- 3.1 Publication rule . . . 6
- 3.2 Masking rule . . . 6
  - 3.2.1 Sea and inland water . . . 7
- 3.3 Static fields and hourly fields . . . 7

4 Atmosphere **8**
- 4.1 Pressure-level variables . . . 8
- 4.2 Below ground . . . 9
- 4.3 The 50 hPa level . . . 9
- 4.4 Near-surface fields . . . 10
- 4.5 Columns and clouds . . . 10
- 4.6 Instability . . . 11

5 Radiation, skin temperature and surface exchanges **13**
- 5.1 Radiation . . . 13
- 5.2 Skin temperature . . . 14
- 5.3 Roughness length, friction velocity and stress . . . 15

6 10 m wind and gust **17**
- 6.1 The `10fg` field . . . 17
- 6.2 Comparison with ERA5 . . . 17
- 6.3 The hourly window . . . 18

7 Water, soil, snow and static fields **19**
- 7.1 Precipitation and runoff . . . 19
- 7.2 Soil . . . 20
- 7.3 Snow . . . 20
- 7.4 Vegetation, soil type and lakes . . . 20
- 7.5 Albedo . . . 21

7.6 Orography 22

8 Limits of the mapping **23**
8.1 Fields computed and not written 23
8.2 Fields excluded by the configuration 24
8.3 Fields excluded by the publication rule 24
8.4 Fields present and not published 24

9 Properties that change the use of the data **26**

A The 90 variables **28**

B Verification method **31**

Bibliography **32**

Chapter 1

The simulation

CHAPTER is a convection-permitting dynamical downscaling of ERA5 over Europe and the Mediterranean basin [1], produced with the WRF (Weather Research and Forecasting) model, version 4.1.1 [2]. The CHAPTER GRIB2 archive is the set of hourly GRIB2 files derived from the wrfout files of the simulation, one per validity time, with the ECMWF parameter names (`shortName`) and codes (`paramId`) (Chapter 2).

1.1 Domain, grid and period

The simulation uses domain d02, a Mercator grid of 1,641 × 1,353 cells with a spacing of 3 km, i.e. 2,220,273 points per field. The standard parallel of the projection is 44.671° N. The domain extends from 23.471,9° to 60.000,6° north latitude and from 19.972,2° W to 42.255,5° E. Vertically the model has 49 levels, with the top at 50 hPa. WRF integrates the equations on a sphere of radius 6,370 km, and the archive declares the same sphere in the grid definition (Section 2.2).

The archive covers, at hourly resolution, the years 2019 and 2024 and the period from 1 January to 2 July 2025, for 21,936 validity times. The last validity time produced by the simulation is 2 July 2025 at 23 UTC. Each daily run starts at 18 UTC on the day before the one it is dedicated to and integrates for 30 h; its first 6 h are spin-up and are not part of the archive. Day D consists of the validity times from 01 to 23 UTC of the run started at 18 UTC on D−1, and its 00 UTC validity time is the last hour, the thirtieth, of the run started at 18 UTC on D−2 (Sections 2.4 and 2.5).

1.2 Physical configuration

Table 1.1 lists the simulation options on which the content of the archive depends, read from the global attributes of the wrfout files. They are the same in the three periods checked: 2019, March 2024 and July 2024.

The radiation scheme is RRTMG, with the CAM ozone climatology (`o3input=2`). The WSM6 microphysics carries five condensed species, among them graupel, which ERA5 does not publish as a separate field (Section 4.5).

In the RUC surface scheme the background albedo, the leaf area index and the roughness length are table values per dominant category, i.e. per land-use class with the largest fraction in the cell. For this reason `al` takes 8 to 11 values over the whole domain and `fsr`, away from snow, one value per category (Chapters 5 and 7).

Turbulent kinetic energy, waves, currents and sea surface height do not exist in this simulation. The boundary-layer height and the sensible and latent heat fluxes, by contrast, were computed but not written to the wrfout files (Chapter 8).

Table 1.1: Simulation options that determine which fields the archive can contain.

Option	Value	Scheme	Consequence for the archive
BL_PBL_PHYSICS	1	YSU	nonlocal scheme, without prognostic turbulent kinetic energy
KM_OPT	4		the diffusion produces no turbulent kinetic energy
SF_OCEAN_PHYSICS	0		ocean not simulated: no currents and no sea surface height
no wave model			no wave fields
SST_UPDATE	0		sea surface temperature constant within each daily run
CU_PHYSICS	0		resolved convection; RAINC identically zero
MP_PHYSICS	6	WSM6	cloud water, rain, ice, snow and graupel; no hail
ICLOUD	1	Xu–Randall	continuous cloud fraction between 0 and 1
SF_SURFACE_PHYSICS	3	RUC	soil on six nodes at 0 cm, 5 cm, 20 cm, 40 cm, 160 cm and 300 cm
SF_LAKE_PHYSICS	0		no lake model
SF_URBAN_PHYSICS	0		no urban scheme
aer_opt	0		radiation without aerosols
usemonalb	.false.		background albedo from a table, not from satellite
rdlai2d	.false.		leaf area index from a table per category
isfflx	1		turbulent surface fluxes computed at every time step
output_diagnostics	0		no hourly extremes of 2 m temperature
nwp_diagnostics	1		hourly maximum of the 10 m wind and six other hourly maxima

1.3 The static file

The static information on soil, vegetation and lakes comes from the static file geo_em.d02, produced by the WRF preprocessor (WPS) for this simulation. Its land-use category (LU_INDEX) coincides with the dominant category written every hour in the wrfout files (IVGTYP) on 100.00 % of the cells in July 2019 and in January 2025, once the reclassification of lakes as water is excluded; the model applies that reclassification because the lake physics is switched off. On 15 February 2025 at 12 UTC the agreement drops to 99.98 %: the 2,735 differing cells all carry sea ice, which the model reclassifies as land at every hour (Section 3.3).

Land use is expressed in 21 MODIS-IGBP classes (MMINLU=MODIFIED_IGBP_MODIS_NOAH). The static file carries the fraction of each class for every cell, whereas the wrfout files keep only the dominant category. On average the dominant category occupies 90.1 % of a land cell, and 17.93 % of land cells have both high and low vegetation above 5 %. The soil texture and the lake depth also come from the static file. The static file made fifteen fields possible: the vegetation covers and types (cvl, cvh, tvl, tvh), the soil type (slt), the lake cover and depth (cl, dl) and the eight soil moisture and temperature fields on the ERA5 layers, to which it supplies the permanent land mask (Chapter 7).

Chapter 2

The archive format

2.1 Files and messages

The archive consists of one GRIB2 file per validity time. Every hourly file contains 90 variables in 246 GRIB messages, always the same ones. The list comes from a single registry in the converter, and every file produced is checked against that list, so a field that could not be computed does not go unnoticed. File names follow the pattern

`ailam-an-cima-3km-{year}-{year}-1h-v1-{YYYYMMDD}{HH}.grib`,

in which the year appears twice with the same value and the final part is the validity time.

The messages use grid definition template 3.10 (Mercator), originating centre `ecmf` and `grid_ccsds` packing at 24 bits per value. Every parameter carries the ECMWF paramId and shortName [3]. There are 89 distinct shortNames for 90 variables, because `z` appears twice, as geopotential on the pressure levels and as surface geopotential, i.e. the orography. Fields defined only over part of the domain carry a GRIB bitmap, and points outside the domain of definition are missing points, not zeros (Section 3.2).

An hourly file is 737 MB on average over the 184 files of September 2024, and 661 MB over the 8 files at 00 UTC among them; over the 476 files of February and March 2024 the mean is 765 MB. This gives between 17.7 GB and 18.4 GB per day and between 6.5 TB and 6.7 TB per year.

2.2 Geometry

The grid declares the shape of the Earth with `shapeOfTheEarth=1` and radius 6,370,000 m, the model sphere (Section 1.1). With this declaration the coordinates that eccodes reconstructs from the grid definition coincide with those of the model to within 2.009 m over the whole domain. Without it, the grid reconstructed by eccodes is displaced by up to about 1.1 km at the northern edge.

2.3 Vertical levels

The 13 three-dimensional variables are on the same 13 pressure levels, 1,000 hPa, 925 hPa, 850 hPa, 700 hPa, 600 hPa, 500 hPa, 400 hPa, 300 hPa, 250 hPa, 200 hPa, 150 hPa, 100 hPa and 50 hPa. These are the levels requested by the LAM AI project, a subset of the 37 pressure levels of ERA5 [4]. Of the others, the 8 above 50 hPa lie above the model top and cannot be produced.

The 246 messages are distributed over the level types of Table 2.1.

Table 2.1: Level types of the 246 messages of an hourly file.

Level type	Messages	Content
`isobaricInhPa`	169	13 variables on 13 pressure levels
`heightAboveGround`	11	wind at 10, 100 and 200 m; temperature, dew-point temperature and relative humidity at 2 m; `10fg`; `vwsh`
`surface`	47	static fields, soil, snow, precipitation, runoff and surface radiation
`entireAtmosphere`	10	column integrals and cloud covers
`nominalTop`	5	radiation at the top of the atmosphere
`mostUnstableParcel`	2	`mucape`, `mucin`
`meanSea`	1	`msl`
`entireLake`	1	`dl`

2.4 Time convention

Instantaneous fields use product definition template 4.0, and their reference time (`dataDate`, `dataTime`) coincides with the validity time. The 19 accumulated fields are total precipitation (`tp`), rain (`tirf`), snowfall (`sf`), the three runoffs (`sro`, `ssro`, `ro`) and the 13 radiative fluxes at the surface and at the top of the atmosphere. They use template 4.8, because the ECMWF definitions of these parameters require the key `typeOfStatisticalProcessing`, which does not exist in template 4.0 [5]. As in ERA5 [4], every accumulated field covers the hour ending at the validity time H; in the archive the reference time is $H-1$ and the accumulation interval (`stepRange`) is 0–1. The value is the difference between two consecutive hours of the same run, for example $\texttt{tp}(H) = [\texttt{RAINNC}(H) - \texttt{RAINNC}(H-1)]/1000$, with `RAINNC` in mm and `tp` in m [6]. The twentieth field with template 4.8 is `10fg`, the maximum of the 10 m wind, with reference time $H-1$, declared interval 0–1 and `stepType=max` (Chapter 6).

The hourly file at 00 UTC of day D comes entirely, accumulated fields included, from the thirtieth hour of the run started at 18 UTC on D−2; the accumulated fields are the difference between its thirtieth and twenty-ninth hours [7, 8]. The accumulated fields of the hourly file at 01 UTC of D are the difference between the seventh and sixth hours of the run started at 18 UTC on D−1. The 24 validity times from 01 UTC of D to 00 UTC of D+1 all come from that run, and the sum of their accumulated fields is the run's total from its sixth to its thirtieth hour. When the wrfout file of a run's thirtieth hour is missing at the source, every message of the following hourly file at 00 UTC consists of missing points; as of 14 October 2026 this happens in no hourly file of the archive.

Since instantaneous fields have reference time H and fields with template 4.8 have $H-1$, an hourly file contains two reference times, and an index of the hourly files must be built on the validity time (`validityDate`, `validityTime`), never on the reference time. Telling an accumulated field from an instantaneous one requires `stepType` (or `typeOfStatisticalProcessing`) and `stepRange`. The key `generatingProcessIdentifier` is 128, the eccodes default for GRIB2, in all 246 messages, and does not separate accumulated fields from instantaneous ones [9].

2.5 The run switch

The *run switch* is the step, once a day, between the 00 UTC validity time of D, thirtieth hour of the run started at 18 UTC on D−2, and the 01 UTC validity time of D, seventh hour of the run started at 18 UTC on D−1. The jump between the two validity times is the difference between the two runs at 00 UTC of D, thirtieth hour of the first and sixth of the second, plus the change over the hour 00–01 UTC in the second. Table 2.2 compares it with one hour of change within the same run, over the full domain, at a single run switch, from 23 UTC on 18 January to 01 UTC on 19 January 2024 [8].

TABLE 2.2: Root-mean-square difference between two consecutive validity times, over the full domain, 18 and 19 January 2024.

Validity times	`2t` (K)	`sp` (hPa)	10 m wind speed (m/s)	`t` at 850 hPa (K)	`tcwv` (kg/m^2)
23–00 UTC, same run	0.66	0.50	1.13	0.48	0.89
00–01 UTC, run switch	1.79	1.62	1.62	0.94	1.54

On 19 January 2024 the root-mean-square difference of `2t` at the run switch is 2.7 times that of one hour within the same run, and the maximum absolute difference between the two validity times is 11.0 K, against 9.1 K between 23 and 00 UTC. A tendency or a time derivative computed across the interval from 00 to 01 UTC contains this jump.

Chapter 3

Rules of correspondence with ERA5

Two rules decide what the archive publishes and where, and they apply to all 90 variables.

3.1 Publication rule

A field is published under an ERA5 name only if the WRF quantity is the one that the ERA5 parameter defines, or if the difference can be stated in one line. A field that meets neither condition is left out of the archive, even when the model computes it. There are two distinct reasons for an exclusion. For some WRF fields no ECMWF parameter describes the same quantity, for others the parameter exists but ERA5 does not publish it (Section 8.4).

Seven fields, in six rows of Table 3.1, are published with a declared difference from their ERA5 namesake.

Table 3.1: Fields published with a declared difference from the ERA5 parameter.

Field	ERA5 parameter	CHAPTER field	Section
`slor`	slope of the sub-grid orography	slope of the terrain resolved at 3 km	7.6
`10fg`	parameterized gust	hourly maximum of the resolved wind at 10 m	6
`iews`, `inss`	components of the IFS turbulent surface stress	ρu_*^2, with ρ the air density and u_* the friction velocity, projected onto the direction of the 10 m wind	5.3
`fsr`	effective roughness length, including the parameterized orographic drag	local roughness length `ZNT`, table value of the dominant category modified by snow	5.3
`lsm`	permanent land–sea mask	hourly WRF mask, which counts sea ice as land	3.3
`tcw`	total column water, five species	sum of six species, graupel included	4.5

3.2 Masking rule

A point is missing where the ERA5 parameter is not defined. A point where the model provides no value, or where the value looks wrong, is not missing for that reason. Missing points are written as a GRIB bitmap, never as zeros, so a reader of the archive can always tell a point where the parameter is not defined from one where it is zero. Table 3.2 lists the masked fields.

The lake cover `cl` is defined over the sea as well, where it is zero, because a sea cell has no lake cover. Four fields are masked for a different reason: at the missing points the WRF quantity has no value that corresponds to the ERA5 parameter. `tvl` is missing over shrubland, because the ERA5 type table separates evergreen and deciduous shrubs and the MODIS classification carries

Table 3.2: Fields with missing points and the domain on which they are defined.

Field	Defined on	Section
`sst`, `ci`	sea cells	3.2.1
`dl`	cells with a lake cover greater than zero (32,000)	7.4
`swvl1-4`, `stl1-4`, `slt`	permanent land (1,304,109 cells)	7.2, 7.4
`tvl`	all cells except the 124,672 where shrubland dominates	7.4
`mucin`	columns with CAPE of at least 100 J/kg	4.6
`tsn`	cells with `SNOWC` above 0.9	7.3
`fal`	points with incoming solar radiation at the surface above 50 W/m^2	7.5

no shrub phenology (Section 7.4). `tsn` is missing where snow does not cover the cell, because there the model field is a soil temperature (Section 7.3). `fal` is missing where the incoming solar radiation at the surface does not exceed 50 W/m^2; without incoming radiation the ratio of reflected to incoming radiation is not defined (Section 7.5). `mucin` is missing where CAPE is below 100 J/kg, because there the routine returns no inhibition (Section 4.6).

3.2.1 Sea and inland water

Masking `sst` and `ci` requires separating the sea from inland water, and the lake class of the static file is not sufficient for this. Class 21 (lake) of the MODIS-IGBP classification includes the Black Sea, the Sea of Azov and part of the Baltic, which together make up 89 % of the 55,881 cells that the static file classifies entirely as lake. Masking on that class would remove `sst` from the Black Sea.

The criterion used is the size of the connected component, i.e. of the set of water cells connected through shared edges. In the domain the largest components have 844,431 cells (Atlantic, Mediterranean, Baltic and North Sea), 49,841 (Black Sea and Sea of Azov) and 8,583 (Red Sea). The next is Lake Vänern, with 1,225 cells, the largest lake in the domain. Components of at least 2,000 cells are sea, for a total of 902,855 sea cells. Any threshold between 1,226 and 8,583 cells gives the same set.

3.3 Static fields and hourly fields

A static field comes from the static file and is therefore identical in every hourly file. The static fields are `cl`, `cvh`, `cvl`, `dl`, `slt`, `tvh` and `tvl`. The soil columns `swvl1-4` and `stl1-4` change every hour, but their mask is the permanent land of the static file.

The land–sea mask `lsm`, by contrast, is the hourly WRF mask (`LANDMASK`), which reclassifies as land every water cell covered by sea ice. The mask therefore grows in winter: in a file of 20 March 2024 the reclassified cells number 2,469, exactly the cells with sea ice. Sea ice `ci` is 0 or 1, because the simulation does not use fractional sea ice (`FRACTIONAL_SEAICE=0`). The permanent land mask, in the ERA5 sense, is the set of points with `lsm` = 1 and `ci` = 0, counting the missing points of `ci` as zero.

The background albedo `al` also changes in time, with sea ice and at the transition between the summer and winter seasons of the land-use table (Section 7.5). For `lsm` and `al` must be read from the hourly file of the validity time, not from an arbitrary file of the archive.

Chapter 4

Atmosphere

4.1 Pressure-level variables

Table 4.1 lists the 13 pressure-level variables, 169 messages in all. Besides temperature, wind, humidity and geopotential, they include the two forms of vertical velocity, in pressure (`w`) and geometric (`wz`), the cloud fraction and four cloud and precipitation water contents.

Table 4.1: Variables on the 13 pressure levels.

Field	paramId	Description	Units
`cc`	248	cloud fraction	0–1
`ciwc`	247	specific cloud ice water content	kg/kg
`clwc`	246	specific cloud liquid water content	kg/kg
`crwc`	75	specific rain water content	kg/kg
`cswc`	76	specific snow water content	kg/kg
`q`	133	specific humidity	kg/kg
`r`	157	relative humidity	%
`t`	130	temperature	K
`u`	131	zonal wind component	m/s
`v`	132	meridional wind component	m/s
`w`	135	vertical velocity in pressure	Pa/s
`wz`	260238	geometric vertical velocity	m/s
`z`	129	geopotential	m^2/s^2

Specific contents. WRF carries vapor and hydrometeors as *mixing ratios*, per kilogram of dry air. ECMWF defines `q` and the water contents as *specific contents*, per kilogram of moist air, and the mass of moist air comprises dry air, vapor, cloud water, ice, rain and snow [3]. The archive therefore divides `q`, `clwc`, `ciwc`, `crwc`, `cswc` and the column integrals by the same denominator, $1 + r_t$, where r_t is the total water mixing ratio. On the three-dimensional field the correction is 0.011 % in the median, 1.44 % at the 99th percentile and 2.26 % at most.

Cloud fraction. With the Xu–Randall scheme the cloud fraction `cc` is continuous. On a subdomain of 500×600 cells only between 18 % and 22 % of the cloudy points are exactly 1, and the others span the whole interval from 0 to 1.

Relative humidity. The field `r` is ECMWF's relative humidity, computed over the mixed phase with equations 7.5, 7.6, 7.89 and 7.90 of the IFS Cy41r2 documentation, the cycle with which ERA5 was produced [4], and with the constants of that cycle [10]. The saturation vapor

pressure is

$$e_{\mathrm{sat}}(T) = \alpha(T)\, e_{\mathrm{sat,w}}(T) + (1-\alpha(T))\, e_{\mathrm{sat,i}}(T), \qquad \alpha(T) = \left(\frac{T - T_{\mathrm{ice}}}{T_0 - T_{\mathrm{ice}}}\right)^2, \tag{4.1}$$

where T is the temperature, $e_{\mathrm{sat,w}}$ and $e_{\mathrm{sat,i}}$ are the saturation vapor pressures over water and over ice (Tetens formula), $T_{\mathrm{ice}} = 250.16\,\mathrm{K}$, $T_0 = 273.16\,\mathrm{K}$, and α is 0 below T_{ice} and 1 above T_0. Below T_{ice} this relative humidity differs by up to a factor of 1.8 from the one computed over liquid water. The field is not truncated at 100 %, because supersaturation is a state of the model and ERA5 publishes it too. In the hourly file of 15 July 2024 at 12 UTC, `r` reaches 139.7 % at 300 hPa in the ice regime, and 34,785 points exceed 100 % at 400 hPa. An ingestion pipeline that discards values above 100 % discards valid data.

Geopotential. The published geopotential is $z = g_0\, z_{\mathrm{WRF}}$, with $g_0 = 9.806{,}65\,\mathrm{m/s^2}$ the standard gravity and z_{WRF} the geopotential height of the model. Dividing `z` by g_0 recovers the model height exactly, at the surface and on every pressure level. The surface geopotential and the pressure-level geopotential follow the same convention, verified point by point on the 1,349,910 land cells with elevation above 1 m on 15 July 2024 at 12 UTC. WRF, however, integrates the hydrostatic equation with $g = 9.81\,\mathrm{m/s^2}$, so the archive departs by 0.034 % from hydrostatic balance with its own `t` and pressure. This is one part in 2,900, about 2 m at 500 hPa and 7 m at 50 hPa [11]. The two properties, `z` divided by g_0 equal to the model height and hydrostatic balance with the model's `t` and pressure, cannot hold together.

4.2 Below ground

Where a pressure level lies below ground, the archive publishes the value of the lowest model level as it is, without extrapolating or filling (*clamping*). Below ground `t` coincides exactly with the lowest model level, `q` within $1 \cdot 10^{-9}$ kg/kg and `z` within $2.5 \cdot 10^{-4}$ m [12]. The value is a real state of the model, taken between 24 m and 27 m above ground, and it stays consistent with the near-surface fields. In July, `2t` − `t`(1,000 hPa) below ground is −0.9 K at midnight and 1.4 K at noon, that is, a nocturnal inversion and a daytime superadiabatic layer in the roughly 22 m between the 2 m level and the lowest model level. Below ground `r` reaches about 163 %, because it pairs the humidity of the lowest model level with the nominal pressure of 1,000 hPa.

The cost is that `z` contradicts `sp`. Where the surface pressure places 1,000 hPa below ground, the geopotential published at 1,000 hPa places it, in the median, between 24 m and 27 m above. This occurs on 47.6 % of the domain at 1,000 hPa and on 9 % at 925 hPa. ERA5 instead extrapolates below ground with a temperature lapse rate, and the two archives differ by convention precisely where 1,000 hPa is below ground. The whole difference at 1,000 hPa comes from this: on the verification sample the bias of `t` against ERA5 is −1.71 K at 1,000 hPa against −0.05 K at 850 hPa, and the correlation of `z` with ERA5 at 1,000 hPa is 0.108. For a comparison with ERA5 or for computing thicknesses, the below-ground levels must be masked with `sp`, which is in the same hourly file.

4.3 The 50 hPa level

The model top is at 50 hPa (Section 1.1), and the highest model level at which WRF defines temperature always lies below it, for example at 51.9 hPa on average over the domain on 19 January 2024 at 00 UTC. The 50 hPa pressure level therefore lies above the highest model level, and the vertical interpolation repeats the values of that level there, as the clamping does below ground (Section 4.2). At that validity time `t` at 50 hPa equals the temperature of the highest model level to within the encoding precision, 0.009 K, and $\mathtt{z}/g_0$ equals its geopotential height to within 1 m; the same holds for the other fields at 50 hPa. At the same validity time the layer

between the highest model level and 50 hPa has a hydrostatic thickness of 227 m on average, between 214 m and 239 m over the domain, and $\mathtt{z}/g_0$ at 50 hPa lies that much below the height of the 50 hPa pressure level. The thickness between 100 hPa and 50 hPa computed from the archive, 4,009 m on average, is therefore about 5.4 % short of the hydrostatic thickness, about 4,236 m. A geostrophic wind computed from `z` at 50 hPa is affected by the horizontal variation of the missing thickness, 25 m between its minimum and maximum over the domain.

4.4 Near-surface fields

Table 4.2 lists the fields at fixed height above ground and the mean sea level pressure.

TABLE 4.2: Fields at fixed height and mean sea level pressure.

Field	paramId	Description	Units	Height
`10u`, `10v`	165, 166	wind components	m/s	10 m
`100u`, `100v`	228246, 228247	wind components	m/s	100 m
`200u`, `200v`	228239, 228240	wind components	m/s	200 m
`10fg`	49	hourly maximum of the resolved wind	m/s	10 m
`2t`	167	temperature	K	2 m
`2d`	168	dewpoint temperature	K	2 m
`2r`	260242	relative humidity	%	2 m
`vwsh`	260068	vertical wind shear between 10 m and 100 m	s^{-1}	100 m
`msl`	151	mean sea level pressure	Pa	sea level

Relative humidity at 2 m. `2r` is the relative humidity over liquid water at every temperature, the World Meteorological Organization convention for screen-level humidity measurements [13], that is, for the observations against which a 2 m humidity is compared. ERA5 does not publish a relative humidity at 2 m. `r` and `2r` therefore follow two different definitions and differ where the temperature is below T_0. A profile that places `2r` at the base of `r`, or a vertical humidity gradient computed between the two, mixes two quantities. Recomputing the relative humidity from `q`, `t` and `sp`, all of which are published, puts both in the same convention.

Dewpoint temperature. The inversion that derives the dewpoint temperature from the vapor gives, at saturated points, a `2d` that exceeds `2t` by 0.013,3 K, the same at every validity time and in every season. At four validity times in January 2024, July 2024 and February 2025 these points number between 6,161 and 64,005, all with a relative humidity of 100 % [14]. The value is left as it is. Truncating `2d` at `2t` would remove the visible crossings but leave the same offset at every unsaturated point, introduce a kink in the field at 100 % and break the consistency between `2d` and the model's 2 m vapor (`Q2`), which holds within a relative error between $3.6 \cdot 10^{-4}$ and $1.1 \cdot 10^{-2}$ on the vapor pressure. A relative humidity reconstructed from `2t` and `2d` therefore reaches 100.09 %.

4.5 Columns and clouds

Table 4.3 lists the column integrals and the cloud covers.

Total water. ECMWF defines `tcw` as the sum of five species, vapor, liquid water, ice, rain and snow [3]. The archive sums six species, because it includes graupel [15]. ECMWF defines a graupel content (paramId 260028, `grle`) and its column (260001, `tcolg`) [3], but ERA5 publishes neither, and graupel appears only inside `tcw`. Reconstructing `tcw` as $\mathtt{tcwv} + \mathtt{tclw} + \mathtt{tciw} + \mathtt{tcrw} + \mathtt{tcsw}$

Table 4.3: Column integrals and cloud covers.

Field	paramId	Description	Units
`tcwv`	137	total column water vapor	kg/m^2
`tclw`	78	total column cloud liquid water	kg/m^2
`tciw`	79	total column cloud ice water	kg/m^2
`tcrw`	228089	total column rain water	kg/m^2
`tcsw`	228090	total column snow water	kg/m^2
`tcw`	136	total column water	kg/m^2
`tcc`	164	total cloud cover	0–1
`hcc`	188	high cloud cover	0–1
`mcc`	187	medium cloud cover	0–1
`lcc`	186	low cloud cover	0–1

does not reproduce it, and the residual is the graupel column. At the four validity times examined, 15 January, April, July and October 2024 at 12 UTC, the residual is, as a domain mean, 0.4 % of the condensate. Its maximum is 5.69 kg/m^2 in January, 13.89 kg/m^2 in April, 16.73 kg/m^2 in October and 26.46 kg/m^2 in July, in a convective cell where it exceeds all the rest of the column's condensate.

Cloud cover. `tcc`, `hcc`, `mcc` and `lcc` are maximum-random overlaps of WRF's cloud fraction `CLDFRA`. On the verification sample, with the CHAPTER fields aggregated onto the ERA5 grid, the ratio of means is 0.783 for total cloud cover and 0.610 for high cloud cover (Table 4.4), and CHAPTER total cloud cover is lower than ERA5's at every one of the 34 validity times [16].

Table 4.4: Cloud covers and total column cloud ice water (kg/m^2): means over the verification sample, on the ERA5 grid.

Field	CHAPTER	ERA5	Ratio of means
`tcc`	0.381	0.487	0.783
`hcc`	0.169	0.276	0.610
`mcc`	0.161	0.205	0.786
`lcc`	0.231	0.268	0.862
`tciw`	0.018,5	0.016,6	1.109

The gap does not come from the overlap assumption. Computed point by point on the archive's `CLDFRA`, the whole range of possible assumptions, from maximum to random overlap, is 2–4 % wide, while ERA5 total cloud cover exceeds CHAPTER's by 28 %, so the assumption accounts for at most one seventh of the gap. The gap lies in `CLDFRA` itself, the Xu–Randall diagnostic. Total column cloud ice is 1.109 times ERA5's while `hcc` is 0.610 times, so the scheme assigns to the condensate present a smaller cloud fraction than ERA5 does. Part of the gap comes from resolution. CHAPTER cloud cover is overlapped at 3 km and then aggregated, whereas ERA5 cloud cover is overlapped on a profile already smooth at 31 km, which for equal layer fractions gives a larger cloud cover. This effect too is bounded by the same range, between 2 % and 4 %.

4.6 Instability

`mucape` (paramId 228235) and `mucin` (228236), in J/kg, are the convective available potential energy (CAPE) and the convective inhibition of the most unstable parcel, that is, the parcel mixed over 500 m with the highest equivalent potential temperature in the lowest 3 km. They are not computed from the surface.

The routine that computes the two quantities returns an inhibition only in columns with CAPE of at least 100 J/kg, and elsewhere `mucin` is a missing point (bitmap). A zero in those columns would state that nothing suppresses convection precisely where it is suppressed most, because a column without CAPE is normally a column with strong inhibition. Where `mucin` is missing, ERA5 carries a mean inhibition between 25 J/kg and 265 J/kg. Filling the missing points with zeros would teach a statistical model that situations with little CAPE have no inhibition, the opposite of the physics.

The fraction of the domain in which `mucin` is present ranges from 3.8 % in January to 28.8 % in July (Table 4.5). Where it is present, `mucin` is between 0.03 and 0.24 times ERA5's, because the columns in which it exists are the least inhibited.

TABLE 4.5: Fraction of the domain in which `mucin` is present, at 12 UTC.

Validity time	Present (%)	Missing (%)
2024-01-15	3.8	96.2
2024-02-15	5.7	94.3
2024-03-15	4.8	95.2
2024-07-15	28.8	71.2
2024-10-15	15.2	84.8
2025-02-15	5.5	94.5

`mucape`, by contrast, is dense, with no missing points: where the routine finds no CAPE, the published zero is the correct value. It is zero at 1,536,450 of the 2,220,273 points (69.2 %) on 15 February 2024 and at 1,020,809 (46.0 %) on 15 July 2024, both at 12 UTC, and at 744,745 (33.5 %) on 4 September 2024 at 15 UTC.

Chapter 5

Radiation, skin temperature and surface exchanges

Over the verification sample the archive carries more downward solar radiation at the surface than ERA5 (1.136 times), less total cloud cover (0.783 times), and a skin temperature over land 1.64 K colder. Radiation, clouds and the surface energy budget must therefore be read together.

5.1 Radiation

The 13 radiative fluxes of Table 5.1 are fields accumulated over the hour ending at the validity time, in J/m^2 (Section 2.4).

Table 5.1: Radiative fluxes, accumulated over the hour ending at the validity time.

Field	paramId	Description
`ssrd`	169	surface downward solar radiation
`ssrdc`	228129	surface downward solar radiation, clear sky
`ssr`	176	surface net solar radiation
`ssrc`	210	surface net solar radiation, clear sky
`strd`	175	surface downward thermal radiation
`strdc`	228130	surface downward thermal radiation, clear sky
`str`	177	surface net thermal radiation
`strc`	211	surface net thermal radiation, clear sky
`tisr`	212	incident solar radiation at the top of the atmosphere
`tsr`	178	net solar radiation at the top of the atmosphere
`tsrc`	208	net solar radiation at the top of the atmosphere, clear sky
`ttr`	179	net thermal radiation at the top of the atmosphere
`ttrc`	209	net thermal radiation at the top of the atmosphere, clear sky

Solar radiation at the surface is higher than in ERA5: the downward flux by 8.5 % under clear sky and by 13.6 % under all-sky conditions (Table 5.2) [17]. The simulation has no aerosol (`aer_opt=0`), whereas ERA5 uses the aerosol climatology of Tegen et al. [18]. The clear-sky flux isolates the cloud-free atmosphere, and the thermal fields, on which aerosol does not act, agree with ERA5 within 1.1 %; the comparison attributes the clear-sky excess to the absence of aerosol. With clouds the excess grows, because the total cloud cover is 0.783 times that of ERA5 (Section 4.5) and lets more radiation through. For solar energy applications, or for training against pyranometer observations, the archive carries a systematic excess of this size, which is a property of the simulation and which the conversion cannot remove.

Table 5.2: Ratio of means between CHAPTER and ERA5 over the verification sample, with the CHAPTER fields aggregated onto the ERA5 grid.

Field	CHAPTER / ERA5
`ssrdc`	1.085
`ssrd`	1.136
`ssrc`	1.116
`ssr`	1.178
`tsrc`	1.031
`strd`	0.989
`strdc`	0.996
`ttr`	1.009
`ttrc`	0.990

5.2 Skin temperature

The simulation did not write the model skin temperature (`TSK`), and the archive reconstructs it as `skt` from the outgoing thermal radiation at the surface. The upward flux $F^{\uparrow}$ (`LWUPB` in WRF) is

$$F^{\uparrow} = \varepsilon \sigma T_s^4 + (1 - \varepsilon) F^{\downarrow}, \tag{5.1}$$

where ε is the surface emissivity, σ the Stefan–Boltzmann constant, T_s the skin temperature and $F^{\downarrow}$ the downward thermal flux at the surface (`GLW`). The emissivity is derived from the simulation itself. RRTMG also computes the clear-sky fluxes, with the same emissivity and the same skin temperature and a different downward flux. Subtracting the two forms of Eq. (5.1) eliminates the temperature:

$$\varepsilon = 1 - \frac{F^{\uparrow} - F^{\uparrow}_{\mathrm{cs}}}{F^{\downarrow} - F^{\downarrow}_{\mathrm{cs}}}, \tag{5.2}$$

where the subscript cs denotes clear sky. The expression is an identity, with no reference temperature and no estimated parameters. Measured in this way, the category emissivity is the same in every cell with the same dominant category, with an interquartile range of zero to the fourth decimal place, and equals 0.980,00 over water. The values per category coincide with the `LEMI` column of the `MODI-RUC` section of `VEGPARM.TBL` [2], from 0.850 on barren soil to 0.980 over water.

With snow the emissivity switches to 0.98 for all categories. The archive applies 0.98 where the snow cover `SNOWC` exceeds the snow-cover threshold, 0.015 (1.5 %), and below the threshold blends the two values. The snow-cover threshold is the only parameter of the reconstruction not read from the simulation: it is the value that minimizes the skin-temperature error in a sweep over the 34 validity times of the verification sample.

The accuracy depends on the season (Table 5.3). Over open water, where the model skin temperature is the sea surface temperature, the reconstruction recovers it with a root-mean-square error of 0.000,5 K on about 910,000 points, at every validity time of the verification sample. Over land the reconstructed emissivity coincides with the model one within 10^{-4} on 99.98 % of the points in July and on 93.05 % in January. The whole of the difference between the seasons comes from the snow rule. Over snow-free land the reconstructed skin temperature differs from the temperature of the soil node at 0 cm (`TSLB`) by a root-mean-square difference of 0.026 K, on 4 September 2024 at 16 UTC.

Against ERA5 the skin temperature over land is 1.64 K colder on average over the 34 validity times of the verification sample. The difference, CHAPTER minus ERA5, ranges from −0.5 K to −1.2 K among the 00 UTC validity times and from −0.8 K to −3.24 K among the 12 UTC ones, with the coldest value on 15 September 2024. Over the sea the difference is 0.01 K, and the correlation with ERA5 is between 0.95 and 0.98 over both land and sea [19]. Over the sea,

00 UTC validity times to be re-measured on the files of the previous day's run (issue 72)

Table 5.3: Accuracy of the reconstructed skin temperature against the model state.

Quantity	Value
open water, root-mean-square error (maximum absolute error)	0.000,5 K (0.002 K)
land, points with emissivity within 10^{-4}	99.98 % in July, 93.05 % in January
land, root-mean-square error in July	0.001,2 K
land, root-mean-square error in January and February	0.084 K to 0.119 K
land, January and February, 99th-percentile absolute error (maximum)	0.81 K (2.79 K)

where CHAPTER and ERA5 use almost the same sea surface temperature, the difference of 0.01 K rules out the reconstruction, the emissivity and the encoding as causes. The error of the reconstruction over land, at most 0.119 K in root-mean-square error, is more than ten times smaller than the mean difference from ERA5.

The difference over land has the opposite sign to the one that radiation and clouds suggest. More solar radiation at the ground and fewer clouds bring energy to the surface, yet the surface is colder, above all at 12 UTC, when the excess of solar radiation is largest. The partitioning of energy at the surface therefore remains unexplained. Explaining it requires the surface energy budget, whose three closing terms, the sensible heat flux, the latent heat flux and the ground heat flux, are not in the archive (Chapter 8). Without these fields four possible causes cannot be separated: the surface scheme may send more energy into latent heat, the soil-moisture climate may differ from that of ERA5, the orography resolved at 3 km raises the mean elevation of a cell aggregated onto the ERA5 grid, or the difference is a model error. Users who compare `skt` with satellite surface temperatures, or correct its bias, must use the measured difference and not an estimate derived from radiation and clouds.

5.3 Roughness length, friction velocity and stress

The roughness length `fsr` is WRF's local roughness length `ZNT`: a table value per dominant category, replaced under snow. It is 0.70 m for forests, 0.15 m for savannas and grasslands, 0.065 m for barren soil, 1.00 m on the 19,867 urban points, and drops to 0.011 m where snow covers the surface. The ERA5 roughness length is instead an effective roughness length, which contains the orographic drag that the IFS parameterizes at 31 km; at 3 km the orography is resolved. Over the sea both roughness lengths follow the Charnock formula. Over land they differ by a factor of two to three (Table 5.4), and a drag coefficient computed from the archive's `fsr` differs from one computed from the ERA5 roughness length through the definition of the roughness length alone [20].

Table 5.4: Roughness length and friction velocity: range of the means per validity time over six validity times spread across the seasons, and ratios to ERA5.

Field	CHAPTER	ERA5	Ratio of means	Ratio of medians
`fsr`, land	0.188–0.217 m	0.375–0.394 m	0.48–0.55	0.32–0.48
`fsr`, sea	0.008,9–0.009,1 m	0.010,3–0.010,5 m	0.85–0.88	1.5–1.9
`zust`, land	0.44–0.52 m/s	0.23–0.28 m/s	1.62–2.06	
`zust`, sea	0.21–0.34 m/s	0.19–0.31 m/s	1.08–1.12	

The friction velocity `zust` is the model's own. The ratio of means to ERA5 is between 1.62 and 2.06 over land and between 1.08 and 1.12 over the sea, where the correlation is between 0.85 and 0.90. A roughness length between 0.32 and 0.55 times that of ERA5, together with a friction velocity between 1.62 and 2.06 times that of ERA5, rules out a simple scale error on the same

roughness length. At 3 km the terrain-forced wind is resolved, and the model friction velocity follows it.

The surface stress components `iews` and `inss`, in $\mathrm{N/m^2}$, are the magnitude ρu_*^2, with ρ the air density and u_* the model friction velocity, projected onto the direction of the 10 m wind. Alignment with the 10 m wind is the only assumption. With τ the vector of components `iews` and `inss`, the identities $|\tau| = \rho u_*^2$ and $u_* = \sqrt{|\tau|/\rho}$ hold within $1.8 \cdot 10^{-6}\,\mathrm{N/m^2}$ and within $2.7 \cdot 10^{-4}\,\mathrm{m/s}$ at every point of six validity times.

Chapter 6

10 m wind and gust

6.1 The `10fg` field

The `10fg` field (paramId 49) is the maximum 10 m wind speed that the simulation produced in the hour preceding the validity time. It comes from `WSPD10MAX`, one of the hourly maxima that WRF writes with `nwp_diagnostics=1`. It is a *resolved wind*: it contains no turbulent term and no estimate of what the grid does not represent [21].

The ERA5 gust is a different quantity. The World Meteorological Organization defines the gust as the maximum of the wind averaged over 3 s, an interval shorter than one time step of a 31 km model, and ERA5 publishes an estimate of it, the *parameterized gust* [3]. In the IFS the parameterized gust is the mean 10 m wind plus a turbulent term proportional to the friction velocity, plus, in convective conditions, a term proportional to the wind shear in the 850–950 hPa layer [10, sec. 3.10.4, eq. 3.99]. The IFS computes it at every time step and archives its maximum since the last output.

The regional simulation receives from ERA5 the state of the atmosphere and of the surface, not the diagnostic fields. The standard ECMWF field table of WPS 4.1 (`Vtable.ECMWF`) contains neither paramId 49 nor 228029 [22]; the table used by this simulation has not been verified. Among the 200 variables of a wrfout file no gust field is present besides `WSPD10MAX`, and the field published as `10fg` is the closest to a gust that the simulation provides.

6.2 Comparison with ERA5

Table 6.1 gives the comparison over 16 validity times. `10fg` is 0.625 times the ERA5 `10fg` and 0.644 times the instantaneous parameterized gust `i10fg`, with correlations of 0.813 and 0.820. Against the instantaneous 10 m wind of the same file it is between 1.004 and 1.032 times that wind, and at none of the 2,220,273 points is it lower. Under paramId 49 the archive publishes the 10 m wind speed, whereas the ERA5 parameterized gust is about 1.6 times its mean wind. The ratio to the ERA5 `10fg` stays between 0.593 and 0.673 at all 34 validity times of the verification sample, in every month and at every hour, and the difference is one of definition.

Table 6.1: The archive's `10fg` against three references, over 16 validity times.

Reference	CHAPTER / reference	Correlation
ERA5 `10fg` (parameterized gust, hourly maximum)	0.625	0.813
ERA5 `i10fg` (parameterized gust, instantaneous)	0.644	0.820
CHAPTER instantaneous 10 m wind, same file	1.004–1.032	

At 3 km part of what ERA5 has to parameterize is resolved, as the friction velocity over land shows, between 1.62 and 2.06 times that of ERA5 (Section 5.3). Read as a gust, `10fg`

underestimates the gust by about 38 %, and no bias correction turns it into a gust, because it is a different quantity. With a correlation of 0.813 with the ERA5 `10fg`, the field is suited to ranking places and hours by windiness and to describing the spatial structure of the wind, and it acts as a lower bound on the gust; it is not suited to estimating the exceedance of a gust threshold.

6.3 The hourly window

The `10fg` messages declare a maximum over the preceding hour (Section 2.4). The hourly ERA5 parameterized gust is also a maximum over the preceding hour, so the length of the window does not explain the difference between the two fields. The encoding is correct. WRF resets `WSPD10MAX` each time it writes a wrfout file, but over six whole days, 15 July 2019 and 1 and 15 February, 20 March, 15 July and 15 August 2024, for 138 transitions from one hour to the next, the reset was missed 17 times, 12.3 % of the cases [23]. In those hours the field contains the maximum over two hours or more, under a message that declares one.

The effect shows in the mean excess over the instantaneous 10 m wind: 0.068–0.311 m/s in the hours with a reset and 0.499–0.784 m/s in those without. On a domain mean of about 4.5 m/s this is an overestimate of 10–17 %. In the hours without a reset the field approaches the ERA5 parameterized gust because the window is longer, and the value alone cannot identify them. Nor do the other fields of the file: the seven WRF hourly maxima are reset by the same event and, when the reset is missed, all carry over together, so none of them signals the missed reset of the others. Recognizing those hours requires the wrfout file of the preceding hour, which is not available at conversion time. The mechanism of the missed reset has not been established.

In statistics over a season, 12.3 % of the hours carry a window longer than the declared one; for a single hour the declared window of `10fg` is not guaranteed.

Chapter 7

Water, soil, snow and static fields

7.1 Precipitation and runoff

Precipitation and runoff are fields accumulated over the hour ending at the validity time (Section 2.4). Total precipitation `tp` (paramId 228, m) is WRF's resolved precipitation `RAINNC`, which is complete because the convective precipitation `RAINC` is identically zero (`CU_PHYSICS=0`, Table 1.1). Snowfall `sf` (144, m of water equivalent) is the sum of resolved snow and graupel, `SNOWNC` + `GRAUPELNC`, that is, all solid precipitation, as in the ECMWF definition [3]. Rainfall `tirf` (235015, kg/m^2) is total precipitation minus snow and graupel; the microphysics has no hail. The three fields form a closed budget. With `tirf` converted to metres, $\texttt{tp} - \texttt{sf} - \texttt{tirf}$ is at most $7.8 \cdot 10^{-8}$ m at the worst point, on 15 February and 15 July 2024 at 12 UTC.

The three runoff fields, surface (`sro`, paramId 8), subsurface (`ssro`, 9) and total (`ro`, 205), carry the information of a single field, and the archive publishes all three [24]. At eight validity times of 2024 (four in Table 7.1), 15 January, February, March and August at 12 UTC and 15 July at 00, 06, 12 and 18 UTC, `ssro` is identically zero at four and, at the others, is non-zero at no more than 17 of the 2,220,273 grid points. At every point of the eight validity times, $|\texttt{ro} - \texttt{sro} - \texttt{ssro}|$ does not exceed $7.5 \cdot 10^{-9}$ m.

Table 7.1: Domain means of the runoff fields, accumulated over the hour ending at the validity time, in m, and number of grid points where `ssro` is non-zero.

Validity time	`sro`	`ssro`	Points `ssro` $\neq 0$	`ro`
2024-02-15 12 UTC	$4.819 \cdot 10^{-6}$	0	0	$4.819 \cdot 10^{-6}$
2024-07-15 12 UTC	$5.543 \cdot 10^{-6}$	$2.2 \cdot 10^{-9}$	3	$5.545 \cdot 10^{-6}$
2024-07-15 18 UTC	$1.829 \cdot 10^{-5}$	$1.5 \cdot 10^{-9}$	2	$1.829 \cdot 10^{-5}$
2024-08-15 12 UTC	$2.334 \cdot 10^{-6}$	0	0	$2.334 \cdot 10^{-6}$

`ssro` is the hourly increment of `UDROFF`, the model's subsurface runoff accumulator. The accumulator is non-zero at 462–8,363 points, with maxima of 11–106 mm. In the run started at 18 UTC on 14 July 2024 the same 1,647 points carry subsurface runoff at its sixth, twelfth, eighteenth and twenty-fourth hours (00, 06, 12 and 18 UTC on 15 July), only 19 of them grow between the sixth and the twenty-fourth hour, and in a single hour no more than 17 grow.

Relative to ERA5, with the accumulations of both datasets summed over the hours from 01 UTC to the validity time, `sro` is 1.0–4.6 times, `ssro` 0.000–0.004 times and `ro` 0.19–0.85 times the ERA5 value. In ERA5 the largest term is subsurface runoff, which in the archive is almost everywhere zero (Table 7.1).

7.2 Soil

Volumetric soil water `swvl1-4` (paramId 39–42, m^3/m^3) and soil temperature `stl1-4` (139, 170, 183, 236, K) are given on the four ERA5 layers, 0–7, 7–28, 28–100 and 100–289 cm. The RUC scheme instead carries point values at six nodes, at 0 cm, 5 cm, 20 cm, 40 cm, 160 cm and 300 cm, with a linear profile between each node and the next. The four ERA5 layers lie between the first and the last node, because ERA5 starts at 0 cm, where the first node sits, and ends at 289 cm, above the node at 300 cm. Each layer is therefore the exact average, over the layer, of the model's linear profile, with no extrapolation at either end. Each layer is a weighted combination of three consecutive nodes, at 0, 5 and 20 cm, at 5, 20 and 40 cm, at 20, 40 and 160 cm and at 40, 160 and 300 cm, with weights that sum to 1. `swvl` uses the total soil moisture, liquid plus frozen, which is what the ERA5 name denotes.

The eight fields are defined on the permanent land of the static file, and sea ice has no soil column, as in ERA5, whose land–sea mask is permanent. Where sea ice forms, WRF reclassifies the cell as land, assigns it soil type 16 and fills the soil column with water: `swvl` reaches exactly 1.000, against a porosity of 0.435 for that soil type. Those columns are a model state but not a soil, and the archive masks them. The ice itself stays in the file, in `ci`.

7.3 Snow

The archive carries the snow water equivalent `sd` (paramId 141, m), the snow density `rsn` (33, kg/m^3), the snow cover `snowc` (260038, %) and the temperature of the snow layer `tsn` (238, K). `tsn` is `SOILT1`, the temperature of the first level of the model's combined snow and soil column, and where snow does not cover the cell that level is soil. The field is therefore defined only where `SNOWC` exceeds 0.9, that is, where `snowc` exceeds 90 %, and is missing elsewhere, including cells with snow on a smaller fraction. The field is missing everywhere at 7 of the 34 validity times of the verification sample, all of them in summer.

7.4 Vegetation, soil type and lakes

Cover. `cvl` (paramId 27) and `cvh` (28) are the sums of the static-file fractions over the MODIS-IGBP classes of low and high vegetation, the same quantity ERA5 denotes (Table 7.2). High vegetation comprises the IGBP classes with tree cover above 30 %, that is, forests (above 60 %) and woody savannas (between 30 and 60 %); savanna, between 10 and 30 %, is low [25].

Vegetation types. `tvl` (29) and `tvh` (30) are, among the low classes and among the high ones, the class with the largest fraction in the cell, translated into WMO code table 4.234 following the correspondence in Table 7.2. The code table distinguishes evergreen shrubs (code 16) from deciduous shrubs (17), whereas MODIS does not record shrub phenology. On the two shrubland classes the code cannot be chosen without an assumption, and `tvl` is missing on the 124,672 cells where shrubland dominates, 16.02 % of the 778,381 cells with `cvl` > 0. `cvl` still reports the cover in those cells.

Soil type. `slt` (paramId 43) does not translate categories. Five of ECMWF's seven soil types are defined by thresholds on the clay and sand percentages [10], and the archive applies them to the clay and sand fractions of the static file (Table 7.3). The thresholds of the coarse, medium and fine types are those of the FAO [26]; those of the medium fine and very fine types have no external source. Once the thresholds are applied, type 6 replaces the result where the dominant soil category is organic material. All 1,304,109 permanent-land cells have a type: 1,069,456 medium, 187,378 coarse, 45,450 fine, 1,498 organic and 327 very fine. Types 3 (medium fine) and 7 (tropical organic) do not occur. eccodes decodes `slt` with WMO code table 4.213, which

Table 7.2: Translation of the MODIS-IGBP classes into WMO 4.234 codes for `tvl` and `tvh`.

MODIS-IGBP class	Set	4.234 code
1 evergreen needleleaf forest	high	3 evergreen needleleaf trees
2 evergreen broadleaf forest	high	6 evergreen broadleaf trees
3 deciduous needleleaf forest	high	4 deciduous needleleaf trees
4 deciduous broadleaf forest	high	5 deciduous broadleaf trees
5 mixed forest	high	18 mixed forest
8 woody savanna	high	19 interrupted forest
9 savanna	low	7 tall grass
10 grassland	low	2 short grass
11 permanent wetland (absent)	low	13 bogs and marshes
12 cropland	low	1 crops, mixed farming
14 cropland and natural vegetation mosaic (absent)	low	1 crops, mixed farming
18, 19, 20 wooded, mixed and barren tundra	low	9 tundra
6, 7 closed and open shrubland	low	none: `tvl` missing
13, 15, 16, 17, 21 urban, snow and ice, barren, water, lake	none	none: fraction $1 - $ `cvl` $-$ `cvh`

has 11 classes and a different meaning; the values of the archive must be read with the ECMWF definition.

Table 7.3: Thresholds for the soil type `slt` (clay and sand in percent).

Condition	ECMWF type
clay ≥ 60	5 very fine
$35 \leq$ clay < 60	4 fine
clay < 35 and sand ≤ 15	3 medium fine
$18 \leq$ clay < 35 and sand > 15, or clay < 18 and $15 <$ sand ≤ 65	2 medium
clay < 18 and sand > 65	1 coarse
dominant soil category of organic material	6 organic

Lakes. The lake cover `cl` (paramId 26) comes from the fraction of the lake class in the static file, and it is non-zero on 32,000 cells. It is not redundant with the hourly land–sea mask `lsm`: 19,309 cells that the mask counts as land contain a lake smaller than the cell. The lake depth `dl` (228007, m) is the `LAKE_DEPTH` field of the static file, and it is not a model state, because the lake physics is switched off (`SF_LAKE_PHYSICS=0`). `dl` is defined on the same 32,000 cells as `cl` and is missing elsewhere; on 22,991 of them it is 10 m, the WPS default value. In the static file `LAKE_DEPTH` takes this default value on 99.56 % of the domain.

7.5 Albedo

The background albedo `al` (paramId 174) is a table value per dominant category. The simulation does not use monthly satellite albedos, and the value comes from the winter and summer columns of the land-use table: the summer column applies from April to October, the winter column from November to March. Over the whole domain the field takes 8 to 11 distinct values, and at the switch between the two columns it changes on 1,171,937 cells, 89.9 % of the permanent-land cells. Each change is one row of the table: for example from 0.20 to 0.17 on cropland on 497,649 cells and from 0.23 to 0.25 on barren soil on 369,219. Over water it is 0.08, over sea ice 0.65. Within a day the value over land does not change; it changes only on sea-ice cells, which follow the

hourly mask (Section 3.3). `al` is neither an observed albedo nor an evolving one, and a difference between two dates on either side of the column switch is not a physical signal.

The effective albedo of the simulation is `fal` (paramId 243), the ratio of reflected to incident solar radiation at the surface. It is defined only where the incident radiation exceeds $50\,\mathrm{W/m^2}$, and is missing elsewhere.

7.6 Orography

The orography is the surface geopotential `z` (paramId 129). The slope `slor` (163) is the magnitude of the gradient of the terrain height h on the 3 km grid, $|\nabla h|$, whereas paramId 163 denotes the slope of the sub-grid orography. eccodes shows the parameter name, *Slope of sub-gridscale orography*, and nothing in the message states that the field is the resolved slope. The standard deviation of orography `sdor` (160), computed by WPS, is the ERA5 quantity, evaluated over a 3 km cell instead of a 31 km one. The ratio of medians against ERA5 is 0.244, equal to $(3/31)^{0.605}$, and the difference comes from the resolution.

Chapter 8

Limits of the mapping

The ERA5 quantities of Table 8.1 have no counterpart in the archive. The limit lies in the wrfout files, which do not contain a field computed by the model, in the configuration of the simulation (Section 1.2) or in the publication rule (Section 3.1).

Table 8.1: ERA5 quantities absent from the archive and origin of the limit.

Parameter	Origin of the limit
boundary layer height (`blh`)	wrfout files: `PBLH` computed and not written
sensible and latent heat fluxes	wrfout files: `HFX` and `LH` computed and not written
turbulent kinetic energy	configuration: no scheme with turbulent kinetic energy
maximum and minimum 2 m temperature (`mx2t`, `mn2t`)	configuration: `output_diagnostics=0`
waves, sea-state-dependent drag coefficient	configuration: no wave model
currents, sea surface height	configuration: ocean not simulated
sea-ice thickness and albedo	configuration: sea ice only as input data
convective precipitation (`cp`, `csf`)	configuration: resolved convection, identically zero field
gust components (`10efg`, `10nfg`)	publication rule
leaf area index per class (`lai_lv`, `lai_hv`)	publication rule
angle and anisotropy of sub-grid orography (`anor`, `isor`)	publication rule
snowmelt (`smlt`)	publication rule
the 8 ERA5 pressure levels above 50 hPa (1 hPa, 2 hPa, 3 hPa, 5 hPa, 7 hPa, 10 hPa, 20 hPa and 30 hPa)	configuration: model top at 50 hPa

8.1 Fields computed and not written

The YSU scheme computes a boundary layer height (`PBLH`), and with `isfflx=1` the sensible and latent heat fluxes (`HFX`, `LH`) are computed at every time step. None of the three was written to the wrfout files, and no more complete output of the simulation exists. The same holds for the model skin temperature `TSK`, which the archive reconstructs from the radiation (Section 5.2).

For the boundary layer height, an a posteriori diagnosis, for example with the bulk Richardson number on the profiles, would be ours and not the model's, and does not satisfy the publication rule (Section 3.1). Nor can the fluxes be reconstructed from the surface energy balance, because the balance keeps three unknowns: the two fluxes and the ground heat flux, which was not written either.

8.2 Fields excluded by the configuration

The YSU scheme is non-local and has no prognostic turbulent kinetic energy, and with `KM_OPT=4` the diffusion does not produce it either: the field does not exist at any height.

With `output_diagnostics=0` WRF did not write the hourly maxima and minima of the 2 m temperature, and an hourly sample of `2t` is not the extreme within the hour. With `nwp_diagnostics=1` the simulation writes hourly maxima, but none for temperature.

No wave model is coupled to the atmosphere, so the significant wave height, the wave period and direction, and the sea-state-dependent drag coefficient are missing. The ocean is not simulated (`SF_OCEAN_PHYSICS=0`), so there are no currents and no sea surface height. Sea ice comes only from the input data, as a 0/1 indicator per cell, with a constant albedo of 0.65; neither the ice thickness nor an evolving ice albedo exists.

With resolved convection (`CU_PHYSICS=0`) the convective precipitation `RAINC` is identically zero, and the ERA5 convective parameters `cp` and `csf` are not published (Section 8.4).

8.3 Fields excluded by the publication rule

Gust components. For `10efg` and `10nfg` only the magnitude of the hourly maximum of the resolved wind exists (Chapter 6), and the two components are excluded by decision.

Leaf area index. The model carries a single leaf area index per cell, a table value for the dominant category (`rdlai2d=.false.`), whereas ERA5 requires one for low vegetation and one for high vegetation. Any split of the single value between the two would be an assumption of ours. The model field `LAI` is in the wrfout files and is not published.

Angle and anisotropy of sub-grid orography. The static file carries the asymmetry (`OA1-4`) and effective length (`OL1-4`) parameters of the Kim–Arakawa scheme. ECMWF's `anor` and `isor` derive from the structure of the covariance of the terrain gradient, and the two quantities are different.

Snowmelt. The field `ACSNOM` exists but is not a monotonic accumulator. On 15 of the 16 days examined some points decrease between 00 and 12 UTC, by up to 8.2 mm, and on 1 July 2024 the decrease relative to 00 UTC reaches 7.26 mm. Built like the other accumulated fields, as the difference between two consecutive hours of the same run (Section 2.4), `smlt` would be negative at those points.

8.4 Fields present and not published

Sixteen fields of the wrfout files are identically zero at every validity time of the verification sample: `ACHFX`, `ACLHF`, `ACGRDFLX`, `NOAHRES`, `SSTSK`, `SST_INPUT`, `SWNORM`, `ACSNOW`, `HAILNC`, `RAINC`, `RAINSH`, `PREC_ACC_C` and the four fields of downward thermal radiation at the top of the atmosphere, `LWDNT`, `LWDNTC`, `ACLWDNT` and `ACLWDNTC`. `ACHFX`, `ACLHF`, `ACGRDFLX` and `NOAHRES` are zero because they are accumulators of the Noah surface scheme, and this simulation uses RUC. An identically zero field is not published, whatever the reason.

For other non-zero fields no ECMWF parameter describes the same quantity: the frozen fraction of precipitation (`SR`), the density of falling snow (`RHOSNF`), the annual maximum and minimum of the green vegetation fraction (`SHDMAX`, `SHDMIN`) and the deep soil temperature used as lower boundary condition (`TMN`). The accumulated runoff `ACRUNOFF` coincides with the surface runoff `SFROFF`, and is already in `sro`.

Five non-zero fields stay out even though ECMWF defines a parameter for the same quantity, because ERA5 does not publish it and the archive follows the ERA5 schema (Table 8.2).

Table 8.2: Non-zero fields of the wrfout files with an ECMWF parameter that ERA5 does not publish.

WRF field	ECMWF parameter
`SNOALB`	260161 `mxsalb`, maximum snow albedo
`QGRAUP`	260028 `grle` and 260001 `tcolg`, graupel and its column (Section 4.5)
`REFL_10CM`	260134 `bref`, radar reflectivity
`LPI`	family 228050 `litoti`; `LPI`, however, is a lightning potential index, not a flash density
`SNOWFALLAC`	3066 `sde`, because it is a geometric snow depth and not a water equivalent

Chapter 9

Properties that change the use of the data

Table 9.1 collects the properties of the archive to be taken into account in a comparison with ERA5 or with observations, each with its measured value and the section that describes it.

Table 9.1: Properties of the archive that change the way it is used.

Property	Value	Section
Downward solar radiation at the surface higher than in ERA5; the simulation has no aerosol	8.5 % in clear sky, 13.6 % with clouds	5.1
Cloud cover lower than in ERA5, because of `CLDFRA` and not of the overlap	ratio of means: `tcc` 0.783, `hcc` 0.610	4.5
Skin temperature over land colder than in ERA5, for causes the archive does not separate	−1.64 K on average, down to −3.24 K at 12 UTC	5.2
`fsr` is WRF's local roughness length, whereas ERA5's is an effective roughness length	over land, ratio of medians 0.32–0.48, six validity times	5.3
`zust` over land responds to the resolved terrain	over land, ratio of means 1.62–2.06, six validity times	5.3
`r` is the relative humidity over the mixed phase, `2r` the one over liquid water	up to a factor of 1.8 below 250.16 K	4.1, 4.4
`r` above 100 % is model supersaturation and is not truncated	up to 139.7 % at 300 hPa on 15 July 2024 at 12 UTC	4.1
The relative humidity reconstructed from `2t` and `2d` exceeds saturation	up to 100.09 %	4.4
Below ground the pressure levels carry the value of the lowest model level (clamping); ERA5 extrapolates	1,000 hPa below ground over 47.6 % of the domain	4.2
`z` is the model geopotential height multiplied by $g_0 = 9.806{,}65\,\mathrm{m/s^2}$; WRF integrates with $g = 9.81\,\mathrm{m/s^2}$	0.034 % from hydrostatic balance	4.1
`10fg` is the resolved wind at 10 m, not a parameterized gust	0.625 times the ERA5 `10fg`, over 16 validity times	6.2
The hourly window of `10fg` is not always one hour	reset missed at 17 of the 138 hourly transitions (12.3 %), over six days	6.3
The three runoff fields carry the information of a single field; drainage at the daily scale is missing	`ssro` non-zero at no more than 17 of the 2,220,273 points, at eight hourly validity times of 2024	7.1
`tcw` sums six species, ECMWF defines five	graupel residual up to 26.46 $\mathrm{kg/m^2}$ at four validity times in 2024	4.5
A point is missing where the ERA5 parameter is not defined; bitmap, never zeros	16 fields with a bitmap (Table 3.2)	3.2

Property	Value	Section
`mucin` is missing where CAPE is below 100 J/kg	from 71.2 % (July) to 96.2 % (January) of the domain, at 12 UTC	4.6
`generatingProcessIdentifier` does not distinguish the accumulated fields	128 in 246 of 246 messages	2.4
`al` is a two-season table value per dominant category	8 to 11 values over the domain	7.5
`slor` is the slope of the resolved terrain under the ECMWF name of the sub-grid slope; `sdor` is computed on 3 km cells	`sdor`: ratio of medians 0.244	7.6

Appendix A

The 90 variables

Table A.1 lists the 90 variables of an hourly file, read from the messages of one hourly file; every file of the archive carries the same 246 messages (Section 2.1). shortName, paramId and units are those returned by eccodes, and the name is that of the ECMWF parameter. The sampling is instantaneous, accumulated over the hour ending at the validity time (Section 2.4), hourly maximum (Chapter 6) or static (Section 3.3). The last column marks the fields with a bitmap of missing points (Section 3.2); w.e. stands for water equivalent.

Table A.1: The 90 variables of the archive.

Field	paramId	ECMWF name	Units	Level	Sampling	Bitmap
cc	248	Fraction of cloud cover	0–1	13 pressure levels	instantaneous	
ciwc	247	Specific cloud ice water content	kg/kg	13 pressure levels	instantaneous	
clwc	246	Specific cloud liquid water content	kg/kg	13 pressure levels	instantaneous	
crwc	75	Specific rain water content	kg/kg	13 pressure levels	instantaneous	
cswc	76	Specific snow water content	kg/kg	13 pressure levels	instantaneous	
q	133	Specific humidity	kg/kg	13 pressure levels	instantaneous	
r	157	Relative humidity	%	13 pressure levels	instantaneous	
t	130	Temperature	K	13 pressure levels	instantaneous	
u	131	U component of wind	m/s	13 pressure levels	instantaneous	
v	132	V component of wind	m/s	13 pressure levels	instantaneous	
w	135	Vertical velocity	Pa/s	13 pressure levels	instantaneous	
wz	260238	Geometric vertical velocity	m/s	13 pressure levels	instantaneous	
z	129	Geopotential	m^2/s^2	13 pressure levels	instantaneous	
100u	228246	100 metre U wind component	m/s	100 m	instantaneous	
100v	228247	100 metre V wind component	m/s	100 m	instantaneous	
10fg	49	Maximum 10 metre wind gust since previous post-processing	m/s	10 m	hourly maximum	
10u	165	10 metre U wind component	m/s	10 m	instantaneous	
10v	166	10 metre V wind component	m/s	10 m	instantaneous	
200u	228239	200 metre U wind component	m/s	200 m	instantaneous	
200v	228240	200 metre V wind component	m/s	200 m	instantaneous	
2d	168	2 metre dewpoint temperature	K	2 m	instantaneous	

Field	paramId	ECMWF name	Units	Level	Sampling	Bitmap
2r	260242	2 metre relative humidity	%	2 m	instantaneous	
2t	167	2 metre temperature	K	2 m	instantaneous	
vwsh	260068	Vertical speed shear	s^{-1}	100 m	instantaneous	
msl	151	Mean sea level pressure	Pa	sea level	instantaneous	
hcc	188	High cloud cover	0–1	column	instantaneous	
lcc	186	Low cloud cover	0–1	column	instantaneous	
mcc	187	Medium cloud cover	0–1	column	instantaneous	
tcc	164	Total cloud cover	0–1	column	instantaneous	
tciw	79	Total column cloud ice water	kg/m^2	column	instantaneous	
tclw	78	Total column cloud liquid water	kg/m^2	column	instantaneous	
tcrw	228089	Total column rain water	kg/m^2	column	instantaneous	
tcsw	228090	Total column snow water	kg/m^2	column	instantaneous	
tcw	136	Total column water	kg/m^2	column	instantaneous	
tcwv	137	Total column vertically-integrated water vapour	kg/m^2	column	instantaneous	
al	174	Albedo (climatological)	0–1	surface	instantaneous	
ci	31	Sea ice area fraction	0–1	surface	instantaneous	yes
cl	26	Lake cover	0–1	surface	static	
cvh	28	High vegetation cover	0–1	surface	static	
cvl	27	Low vegetation cover	0–1	surface	static	
fal	243	Forecast albedo	0–1	surface	instantaneous	yes
fsr	244	Forecast surface roughness	m	surface	instantaneous	
iews	229	Instantaneous eastward turbulent surface stress	N/m^2	surface	instantaneous	
inss	230	Instantaneous northward turbulent surface stress	N/m^2	surface	instantaneous	
lsm	172	Land-sea mask	0–1	surface	instantaneous	
ro	205	Runoff	m	surface	accumulated	
rsn	33	Snow density	kg/m^3	surface	instantaneous	
sd	141	Snow depth	m w.e.	surface	instantaneous	
sdor	160	Standard deviation of sub-gridscale orography	m	surface	instantaneous	
sf	144	Snowfall	m w.e.	surface	accumulated	
skt	235	Skin temperature	K	surface	instantaneous	
slor	163	Slope of sub-gridscale orography	–	surface	instantaneous	
slt	43	Soil type	code	surface	static	yes
snowc	260038	Snow cover	%	surface	instantaneous	
sp	134	Surface pressure	Pa	surface	instantaneous	
src	198	Skin reservoir content	m w.e.	surface	instantaneous	
sro	8	Surface runoff	m	surface	accumulated	
ssr	176	Surface net short-wave (solar) radiation	J/m^2	surface	accumulated	
ssrc	210	Surface net short-wave (solar) radiation, clear sky	J/m^2	surface	accumulated	
ssrd	169	Surface short-wave (solar) radiation downwards	J/m^2	surface	accumulated	
ssrdc	228129	Surface short-wave (solar) radiation downward clear-sky	J/m^2	surface	accumulated	
ssro	9	Sub-surface runoff	m	surface	accumulated	
sst	34	Sea surface temperature	K	surface	instantaneous	yes
stl1	139	Soil temperature level 1	K	surface	instantaneous	yes
stl2	170	Soil temperature level 2	K	surface	instantaneous	yes
stl3	183	Soil temperature level 3	K	surface	instantaneous	yes
stl4	236	Soil temperature level 4	K	surface	instantaneous	yes
str	177	Surface net long-wave (thermal) radiation	J/m^2	surface	accumulated	
strc	211	Surface net long-wave (thermal) radiation, clear sky	J/m^2	surface	accumulated	
strd	175	Surface long-wave (thermal) radiation downwards	J/m^2	surface	accumulated	

Field	paramId	ECMWF name	Units	Level	Sampling	Bitmap
strdc	228130	Surface long-wave (thermal) radiation downward clear-sky	J/m^2	surface	accumulated	
swvl1	39	Volumetric soil water layer 1	m^3/m^3	surface	instantaneous	yes
swvl2	40	Volumetric soil water layer 2	m^3/m^3	surface	instantaneous	yes
swvl3	41	Volumetric soil water layer 3	m^3/m^3	surface	instantaneous	yes
swvl4	42	Volumetric soil water layer 4	m^3/m^3	surface	instantaneous	yes
tirf	235015	Time integral of rain flux	kg/m^2	surface	accumulated	
tp	228	Total precipitation	m	surface	accumulated	
tsn	238	Temperature of snow layer	K	surface	instantaneous	yes
tvh	30	Type of high vegetation	code 4.234	surface	static	
tvl	29	Type of low vegetation	code 4.234	surface	static	yes
z	129	Geopotential	m^2/s^2	surface	instantaneous	
zust	228003	Friction velocity	m/s	surface	instantaneous	
dl	228007	Lake total depth	m	lake	static	yes
tisr	212	TOA incident short-wave (solar) radiation	J/m^2	top of the atmosphere	accumulated	
tsr	178	Top net short-wave (solar) radiation	J/m^2	top of the atmosphere	accumulated	
tsrc	208	Top net short-wave (solar) radiation, clear sky	J/m^2	top of the atmosphere	accumulated	
ttr	179	Top net long-wave (thermal) radiation	J/m^2	top of the atmosphere	accumulated	
ttrc	209	Top net long-wave (thermal) radiation, clear sky	J/m^2	top of the atmosphere	accumulated	
mucape	228235	Most-unstable CAPE	J/kg	most unstable parcel	instantaneous	
mucin	228236	Most-unstable CIN	J/kg	most unstable parcel	instantaneous	yes

Appendix B

Verification method

Every hourly file is checked against the list of 246 messages before it enters the archive (Section 2.1). The properties of Chapters 4 to 7 were measured on a verification sample of 34 validity times: 4 in 2019, 26 in 2024, covering every month, and 4 in 2025. Of these, 16 are at 00 UTC, 16 at 12 UTC, one at 06 UTC and one at 18 UTC. Where a number is measured on a different set, the text says so.

The 16 validity times at 00 UTC were measured on the sixth hour of the same day's run; the archive now takes them from the thirtieth hour of the previous day's run (issue 72): to be re-measured

The comparisons with ERA5 use the CHAPTER fields aggregated onto the ERA5 grid with an area-weighted mean. A ratio between CHAPTER and ERA5 is the ratio of means, that is, the mean of the CHAPTER field divided by the mean of the ERA5 field over the same population of points, or, where stated, the ratio of medians.

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